

KRASNOYARSK MININO, Russian Federation

WMO: 295700

Lat: 56.067N

Lon: 92.733E

Elev: 235

StdP: 98.54

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.6	(c) -33.8	(d) -39.0	(e) 0.1	(f) -36.5	(g) -36.2	(h) 0.1	(i) -33.7	(j) 8.5	(k) -9.8	(l) 7.9	(m) -8.8	(n) 0.4	(o) 230	(p) 0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 29.3	(d) 19.2	(e) 27.5	(f) 18.5	(g) 25.7	(h) 17.7	(i) 20.7	(j) 27.0	(k) 19.6	(l) 25.4	(m) 18.7	(n) 23.9	(o) 2.2	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.8	23.5	17.6	13.0	22.4	16.6	12.2	21.6	61.2	26.8	57.4	25.5	54.3	24.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.2	(b) 6.2	(c) 5.3	(e) -36.9	(f) 32.8	(g) 4.2	(h) 1.8	(i) -39.9	(j) 34.1	(k) -42.4	(l) 35.1	(m) -44.8	(n) 36.1	(o) -47.8	(p) 37.4		
			DB													
			WB													
				-37.0	22.2	4.1	1.2	-40.0	23.0	-42.4	23.8	-44.7	24.4	-47.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-16.9	-13.0	-4.7	3.5	10.0	17.3	18.9	16.1	9.3	2.0	-8.0	-13.6
	DBStd	13.88	9.77	8.31	6.44	5.34	5.51	3.98	3.07	3.70	4.50	5.66	8.33	8.82
	HDD10.0	3804	833	644	455	204	71	3	0	3	66	252	540	733
	HDD18.3	6133	1091	878	713	446	264	65	29	88	271	506	790	992
	CDD10.0	819	0	0	0	8	71	221	277	193	46	4	0	0
	CDD18.3	107	0	0	0	0	5	33	49	19	1	0	0	0
	CDH23.3	1001	0	0	0	1	62	364	396	163	15	0	0	0
	CDH26.7	245	0	0	0	0	13	93	98	39	2	0	0	0
Wind	WSAvg	2.2	2.2	2.2	2.4	2.6	2.5	1.9	1.7	1.8	2.0	2.4	2.4	2.3
Precipitation	PrecAvg	445	16	14	17	24	41	62	61	68	47	36	30	27
	PrecMax	553	32	30	37	46	86	117	116	122	97	66	63	55
	PrecMin	335	1	3	0	3	7	24	2	17	11	11	11	1
	PrecStd	57	7	7	10	12	24	23	28	26	21	14	13	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	4.1	11.2	21.2	28.6	31.6	31.5	30.4	25.5	17.6	10.6	3.0
		MCWB	-0.5	0.2	4.9	10.6	15.6	19.2	20.9	20.7	16.4	10.1	6.0	-0.5
	2%	DB	-0.6	1.7	8.0	17.4	24.9	29.2	29.3	27.4	21.6	14.7	7.0	0.6
		MCWB	-2.6	-1.6	2.9	8.8	13.9	18.8	19.7	19.3	14.5	9.1	3.4	-1.6
	5%	DB	-2.3	0.0	5.9	14.7	22.3	27.3	27.3	24.9	19.0	12.2	4.6	-1.0
		MCWB	-4.3	-2.4	1.8	7.8	13.1	17.8	19.3	18.2	13.3	7.9	1.6	-2.7
	10%	DB	-4.1	-1.9	3.8	12.3	19.8	25.3	25.5	22.8	16.5	9.9	2.4	-2.6
		MCWB	-5.8	-3.9	0.4	6.4	11.7	17.0	18.4	17.0	12.0	6.4	0.3	-4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.7	5.4	11.4	17.0	21.2	22.6	21.7	17.1	11.5	6.5	0.1
		MCDB	1.2	3.9	10.1	19.0	26.2	28.6	28.5	28.0	23.2	15.8	9.8	1.7
	2%	WB	-2.4	-1.1	3.5	9.7	15.2	20.0	21.0	19.9	15.1	9.7	3.7	-1.4
		MCDB	-0.6	1.2	7.4	16.2	22.8	27.0	27.0	25.6	20.7	13.8	6.7	0.5
	5%	WB	-4.3	-2.5	2.0	8.1	13.7	19.0	20.0	18.7	13.8	8.1	1.7	-2.6
		MCDB	-2.5	-0.2	5.4	14.1	20.7	25.1	25.5	23.7	17.9	11.8	4.1	-1.0
	10%	WB	-5.8	-4.0	0.6	6.5	12.2	17.8	19.1	17.6	12.4	6.6	0.3	-4.2
		MCDB	-4.1	-1.9	3.6	12.1	18.9	23.7	24.2	22.1	16.1	9.8	2.3	-2.6
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.2	9.3	9.7	10.6	11.1	9.9	9.6	8.9	7.2	6.6	6.9
		MCDBR	6.7	8.4	11.2	14.3	14.7	15.2	13.3	13.8	13.7	10.8	6.7	7.1
		MCWBR	5.8	6.6	7.6	7.8	7.4	6.6	5.7	6.5	7.2	6.6	4.5	6.0
	5% WB	MCDBR	6.7	7.8	10.4	13.2	13.2	12.6	11.6	12.1	12.4	10.1	6.6	7.3
		MCWBR	6.0	6.4	7.3	7.7	7.0	6.1	5.6	6.1	6.9	6.4	4.7	6.5
Clear-Sky Solar Irradiance	taub		0.284	0.317	0.364	0.396	0.414	0.410	0.433	0.414	0.371	0.352	0.319	0.285
	taud		2.025	2.006	1.966	2.128	2.192	2.300	2.249	2.270	2.339	2.237	2.065	1.971
	Ebn at Noon		578	710	770	814	825	834	803	793	777	677	521	455
	Edh at Noon		65	100	134	133	133	121	125	115	93	80	63	54
All-Sky Solar Radiation	RadAvg		0.80	1.78	3.13	4.30	4.93	5.94	5.29	4.24	2.73	1.44	0.72	0.47
	RadStd		0.09	0.12	0.24	0.28	0.52	0.49	0.36	0.37	0.36	0.11	0.10	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page